

FFGFT $\leftrightarrow$ HLV: the bridge and the fork

A Z_3 circulant from Krüger's C_3 axis, separated by 5- vs 3-fold symmetry

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Abstract

Marcel Krüger's HLV builds space as a 6D→3D icosahedral cut-and-project quasicrystal. Whether FFGFT and HLV are connected is tested here strictly against the three-part criterion: (1) the same structural object from each model's *own* geometry, (2) an explicit map between the derivations, (3) a divergence point that rules out mere Stufe-0 scaffolding. Finding: a Z_3 circulant – the type of FFGFT's mass operator – demonstrably lives inside HLV's own geometry, along a C_3 axis of the icosahedral group (criteria 1+2 met). But the internal invariant diverges *fundamentally*: HLV's protected structure rigidly gives $r = 2/\sqrt{5}$ (5-fold, φ), while FFGFT requires $r = \sqrt{2}$ (3-fold, Koide $Q = 2/3$). No cut-and-project parameter reaches $\sqrt{2}$ ($r(\tau) \leq 1$ for all τ); $\sqrt{2}$ appears in HLV only as an unprotected accidental angle. Result: a *genuine* connection with a *fundamental* fork at the symmetry order – no Stufe-0 triviality, no identity. The fork sits in *space*: HLV's phase layer is its spiral-time operator (not $\theta = 0$), and at the time/memory level HLV's χ kernel and FFGFT's Doc. 282 kernel are non-Markovian kernels of the same class – a bridge candidate (§ 8). All geometric statements are reproducible with `ffgft_hlv_c3_bridge_probe.py`.

1 Occasion and criterion

The exchange with Marcel Krüger (HLV/HICE) repeatedly raises the question whether FFGFT and HLV are connected. An honest answer needs a sharp criterion, or one ends up with pseudo-connections that merely share scaffolding (both use Planck units, both have a square root somewhere). The criterion has three parts:

1. **Same object:** the same structural object derived from each model's *own* geometry – not from shared scale scaffolding.
2. **Map:** an explicit map between the two derivations.
3. **Divergence point:** a point where the models separate – otherwise the "connection" is either trivial identity or Stufe-0 scaffolding.

This document applies the criterion concretely. It claims no theoretical equivalence; it locates precisely *where* FFGFT and HLV meet and *where* they diverge.

2 How Krüger builds his space

The construction is read directly from the HLV homology script (Doc. 271/276): a **6D→3D icosahedral cut-and-project quasicrystal**. A $\mathbb{Z}^6$ lattice is split by two projectors into a 3D parallel space (physical) and a 3D perpendicular space (the "hidden layer"); an acceptance window in perp space selects which lattice points project into the physical quasicrystal. Krüger's exact projector:

$$P(\tau) = \begin{pmatrix} 1 & -1 & 0 & 0 & \tau & \tau \\ \tau & \tau & 1 & -1 & 0 & 0 \\ 0 & 0 & \tau & \tau & 1 & -1 \end{pmatrix} \quad (\text{columns normalised}),$$

with $\tau = \varphi$ (golden) in parallel space and $\tau = \sigma = -1/\varphi$ in perp space. The six columns are the six icosahedral five-fold axes (one per antipodal pair). Internal symmetry: icosahedral (5-fold); the point set is aperiodic. Striking for a bridge: the hidden space is 3D – the same dimension count as FFGFT's generation fibre $\mathbb{C}^3$.

.3 The shared object: a Z_3 circulant on the C_3 axis

The icosahedral group contains C_3 axes. The cyclic coordinate map $(x, y, z) \mapsto (z, x, y)$ is an order-3 rotation; it splits Krüger's six columns into two 3-cycles, $\{c_1, c_3, c_5\}$ and $\{c_2, c_4, c_6\}$. The Gram matrix of such a C_3 orbit is – forced by the C_3 symmetry – an *exact real Z_3 circulant*, precisely the type of FFGFT's mass operator $S = \text{circ}(t_0, t_1, t_2)$ on $\mathbb{C}^3$:

$$G_{\text{orbit}} = \text{circ}\left(1, \frac{1}{\sqrt{5}}, \frac{1}{\sqrt{5}}\right) \quad (\tau = \varphi, \text{ normalised}),$$

with $r = 2g_1/g_0 = 2/\sqrt{5} \approx 0.894$ and phase $\theta = 0$ (real). The perp space ($\tau = \sigma$) gives $r = -2/\sqrt{5}$. Closed form: $|r| = 2|\tau|/(1 + \tau^2)$, and for $\tau = \varphi$ the φ cancels exactly to $2/\sqrt{5}$ ($1 + \varphi^2 = \sqrt{5}\varphi$).

Criteria 1 and 2 met

A Z_3 circulant lives *inside* HLV's own icosahedral geometry – not imposed, but as the Gram matrix of a C_3 orbit. This is the same object as FFGFT's mass operator (criterion 1). The map is explicit (criterion 2): FFGFT's Z_3 generation symmetry $\leftrightarrow$ a C_3 axis of HLV's icosahedral group, with the orbit $\{c_1, c_3, c_5\}$ as image.

.4 The fork: 5-fold versus 3-fold

The divergence point (criterion 3) is sharp and, as three checks show, *fundamental* – not closable by any parameter.

Only $1/\sqrt{5}$. All 6×6 inner products of Krüger's columns have *a single* magnitude: $1/\sqrt{5}$. The 5-fold geometry knows only the angle $\arccos(1/\sqrt{5}) \approx 63.4^\circ$; the 45° angle ($\cos = 1/\sqrt{2}$) that FFGFT needs does not occur between the vacuum directions. Every protected 3-orbit therefore gives $r = 2/\sqrt{5}$.

No τ reaches $\sqrt{2}$. The closed form $r(\tau) = 2\tau/(1 + \tau^2)$ is capped by 1 for *every* τ (maximum at $\tau = 1$, cubic). FFGFT requires $r = \sqrt{2} \approx 1.414 > 1$ – which is *outside* the reachable range of Krüger's construction, for any parameter.

$\sqrt{2}$ needs 3-fold. In a general 3-fold geometry (cyclic basis (a, b, c) , $r = 2(ab + bc + ca)/(a^2 + b^2 + c^2)$) $\sqrt{2}$ is perfectly reachable, e.g. $v = (1, 1, \sqrt{2} - 2^{1/4})$. FFGFT's Koide point ($\text{circ}(1, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}})$, eigenvalues 0.29/0.29/2.41, positive semidefinite) is thus geometrically legitimate – it merely needs a *3-fold* column, not Krüger's 5-fold one.

	r	θ	Koide Q	source
HLV (protected)	$2/\sqrt{5} = 0.894$	0	7/15	5-fold / φ
FFGFT (mass op.)	$\sqrt{2} = 1.414$	$2/9$	$2/3$	3-fold / Koide

.5 Protected versus generic: no loophole

One objection remains: the HLV quasicrystal has thousands of projected points – could a C_3 orbit of *some* point accidentally hit $\sqrt{2}$? For 9822 projected points in the acceptance window the C_3 -orbit ratios g_1/g_0 do fill the whole continuum $[-\frac{1}{2}, 1]$, and $1/\sqrt{2}$ is among them 150 times.

A generic hit proves nothing

That generic projected points sit at *every* angle – including $1/\sqrt{2}$ – holds for any dense 3-fold point set, even a cubic one. This is the Stufe-0 triviality: one picks an *unprotected* direction with the right angle. The *symmetry-protected* orbits – Krüger's six vacuum directions – by contrast rigidly give only $\pm 1/\sqrt{5}$. $\sqrt{2}$ is not present in HLV's protected structure; it appears only as a meaningless accidental angle.

.6 The bridge is containment, not contradiction

The fork above is a statement about the *protected* invariant; it must not be read as the two models being incompatible realities. At the level of groups and dimensions they are reconcilable by *containment*. The icosahedral group contains the 3-fold: $C_3 < A_5$ – a 120° rotation about a $(1, 1, 1)$ axis and a 72° vertex rotation are *both* symmetries of the same icosahedron – and $T^7 = T^4 \times T^3$ ($7 = 4 + 3$). Hence $HLV \supset FFGFT$: HLV is FFGFT's 4D core extended by the three perp circles and the full 5-fold. What the fork establishes is therefore not incompatibility but the *realised protected geometry*: FFGFT protects an independent 3-fold column ($\sqrt{2}$, Koide $2/3$), HLV protects the 5-fold ($2/\sqrt{5}$), and $\sqrt{2}$ is not among HLV's protected directions (it appears only as an unprotected accidental angle, previous section). The models share the ambient $3+1$ and are group/dimension-reconcilable while realising different protected structure – which is the empirically decidable difference, not a coordinate artifact and not a logical incompatibility. Computed in Doc. 285 (`recompute_reconcile.py`, `dim_bridge_1..3`).

Where this is asserted. All of this lives in the compactified picture — the tori with every circle compact, where no circle is singled out as time. The bridge to physical, decompactified observables runs through the single step $S^1 \rightarrow \mathbb{R}$, the measurement (Doc. 270); the containment is therefore asserted upstream of the measurement, in the compact state, not on the physical-time axis.

.7 HLV is the decompactified variant: mass and time only through the addition

One layer of the fork sits deeper than the symmetry order and is easy to miss. HLV's cut-and-project maps into an infinite, aperiodic $\mathbb{R}^3$ – it is the *decompactified* variant. FFGFT's space is by contrast *compact* (T^4). This is not cosmetic: mass and time are intrinsic in FFGFT because compactness quantises the modes – that is the time-mass duality $\tilde{T} \cdot m = 1$. Without compactness there is no mode quantisation, hence *no intrinsic mass scale and no time*. That is exactly why HLV's protected structure is purely geometric: a real Gram metric with $\theta = 0$ and dimensionless angles. The intrinsic *mass* scale is missing because the decompactified variant has no mode quantisation; it arrives, as already recorded in Doc. 282, only *through the addition*. But it matters that this $\theta = 0$ is the spatial metric, *not* HLV's time. HLV is not timeless or phaseless – its time sits in the native spiral-time operators (§ 8), which the spatial Gram does not capture at all. The C_3 probe saw only the geometry, not the dynamics.

This sharpens criterion 1. The shared Z_3 circulant is the same only *algebraically* (the same cyclic form). Physically it differs: in HLV (decompactified) a *spatial metric* without mass and time, in FFGFT (compact) the *mass operator* with mass and time intrinsic. The bridge carries the form, not the content.

The fork has three layers

layer	HLV	FFGFT
compactness	decompactified (metric)	compact ($\tilde{T} \cdot m = 1$)
symmetry order	5-fold ($\varphi, 1/\sqrt{5}$)	3-fold ($\sqrt{2}$)
phase	spiral-time: skew-adj. R (dyn., § 8)	scalar, $\theta = 2/9$

The complex holonomy ($\theta = 2/9$, FFGFT) demanded above is the *operational consequence* of compactness: there mass and phase arise from the quantised modes. The HLV column of the phase row is subtler than a bare $\theta = 0$ – that was the *spatial* Gram metric; HLV's own time-phase sits in the skew-adjoint spiral-time operator R (§ 8). A full recomcompactification of HLV must *co-integrate* these operators, not merely periodise the space – and a compact HLV is moreover no longer a quasicrystal.

.8 A compactified variant of HLV: the rational approximant

Can this recomcompactification be carried out concretely, to make the comparison fair? Yes – with an established notion from quasicrystal physics, the *rational approximant*. One replaces the irrational φ in the projector by rational Fibonacci ratios $\tau_n = F_{n+1}/F_n$ ($1, 2, \frac{3}{2}, \frac{5}{3}, \frac{8}{5}, \dots \rightarrow \varphi$). For rational $\tau = p/q$ the physical projector, multiplied by q , becomes *integer* – the projection of $\mathbb{Z}^6$ lands on a scaled integer lattice $(1/q)\mathbb{Z}^3$, i.e. on a *periodic crystal* on T^3 . For irrational φ no such factor exists; the point set stays dense and aperiodic. That is the precise sense of “compactified HLV” – and, as § 6 already noted, it is no longer a quasicrystal but its periodic crystal approximant.

The approximant is periodic and therefore has discrete modes – a *mass scale becomes possible*. This closes the deepest layer of the fork (compactness): HLV is now in the same regime as FFGFT. The C_3 ratio converges:

$\tau = F_{n+1}/F_n$	$r(C_3)$	Koide Q
2/1	0.800	0.440
5/3	0.882	0.463
13/8	0.893	0.466
$\rightarrow \varphi$	$0.894 = 2/\sqrt{5}$	7/15

But the approximant inherits HLV's 5-fold symmetry: $r \rightarrow 2/\sqrt{5}$, Koide $\rightarrow 7/15$; the *spatial* metric stays real ($\theta = 0$). So the symmetry order keeps diverging. The phase layer, however, is not settled by “ $\theta = 0$ ”: the purely spatial approximant only recomcompactifies space; HLV's time-phase sits in the spiral-time operators it still has to integrate (§ 8).

The fork survives the fair comparison

With the approximant the comparison is *compact vs compact*, and thus fair. It confirms the fork: *even compactified*, HLV gives Koide 7/15, not FFGFT's 2/3. The divergence was therefore *not a compact/decompact artefact* – it sits genuinely in the 5- vs 3-fold symmetry. One can compactify HLV; the finding holds.

.9 What compactification must integrate: HLV's spiral-time

The approximant of § 7 recompactifies HLV's *space*. But HLV is not purely spatial: its defining dynamics are the native 6D *spiral-time* operators

$$\Psi(t) = t + i\phi(t) + j\chi(t), \quad (1)$$

with $\phi(t)$ a phase/coherence channel and $\chi(t)$ a memory/hysteresis channel (Krüger, Zenodo 20668125 and 20512650). A compactified variant that is to *remain* HLV must integrate these operators – the purely spatial approximant does not yet do so. This corrects the phase layer of the fork.

Krüger does not apply Ψ directly (that would give hypercomplex fields) but via a *conservative translation* into two real operators:

channel	becomes	requirement
ϕ (phase)	phase/vorticity ordering	skew-adjoint generator R (energy-neutral)
χ (memory)	history dependence	positive-type kernel K (dissipative)

with exponential example $K_\tau(t) = \frac{1}{\tau}e^{-t/\tau}$, which reduces to an internal variable. Three things follow:

1. **The phase layer is not $\theta = 0$.** The $\theta = 0$ of § 3 is the *spatial* Gram metric, not HLV's time. HLV's phase sits in the skew-adjoint operator R , dynamically driven by $\dot{\phi}(t)$. So the layer reads correctly: *dynamic operator-phase R* (HLV) vs *fixed scalar phase $\theta = 2/9$* (FFGFT) – not phaseless vs phased.
2. **The time/memory layer is a real bridge candidate.** HLV's χ channel is a positive-type, non-Markovian memory kernel $\int_0^t K(t-s)u(s)ds$. FFGFT's time extension in Doc. 282 gives a memory kernel as the Fourier transform of the discrete spectral density $\sum_k g_k^2 \delta(\omega - \omega_k)$ – non-Markovian with *back-flow and revivals*, hence a *discrete-spectrum, oscillatory* kernel (exponential limit $\rightarrow$ internal variable). This is not quite the same shape as HLV's χ in the fluid sector, which is a *positive decaying* kernel (Krüger's Navier-Stokes extension, Zenodo 20512650): both non-Markovian and positive-type, but of different form. Where space forks (5- vs 3-fold), at the time level two kernels of the same *class* (non-Markovian positive memory) face each other: *a spatial fork, a possible χ -memory bridge*.
3. **Two honest limits.** First: the $i \cdot j$ product relation of the spiral algebra is fixed in *none* of Krüger's sources (Doc 271, FA-Hub, Navier-Stokes) – it is always bypassed via the operator translation. The phase comparison must therefore be made at the *operator* level, not the algebraic one; whether operator-phase R meets FFGFT's $2/9$ can only be decided with an explicit R specification. Second: Krüger himself records that many ordinary reduced models generate memory; the HLV reading is only a candidate model class, not a demonstrated ontology (Run M1: the HLV kernel beats weak nulls but, on the *overdamped* observable sector, is statistically *indistinguishable* from a generic biexponential – HLV favoured only by parsimony/BIC). The χ bridge is thus *structural kinship*, not a uniqueness claim – consistent with Doc. 276/281 (φ is not specific either). This is sector-dependent: in the *coherence/revival sector* FFGFT's discrete-spectrum kernel carries a signature (BLP back-flow 5.125) that any *decaying* kernel misses – though uniqueness against other *structured* kernels stays open.

The three levels are not equally strong

Reading the connection by physical level – space, time, mass – the three are *not* equivalent, and this is methodologically decisive:

- **Mass (Z_3 circulant):** the *genuine* bridge – derived from each geometry on its own, with an explicit C_3 map and a real divergence point. It alone fully passes the three-part criterion.
- **Time (χ memory):** *structural kinship*, a candidate – not HLV-unique (Run M1: on the overdamped sector a generic biexponential is *indistinguishable*; HLV favoured only by parsimony). A bridge candidate, not a uniqueness claim.
- **Space (connection-Laplace on a fibre bundle):** *shared language*, not a bridge. That both are a connection-Laplace on a fibre-valued space holds for *every* lattice gauge theory – this is Stufe-0 scaffolding, exactly what the criterion excludes. It must not be counted as a load-bearing connection, or one inflates the link toward triviality.

So only the mass level is a bridge in the sense of the criterion; the time level an honest candidate; the space level a shared frame, no more.

.10 What remains: hexagonal, not pentagonal

Strip HLV of the two axes along which it forks – symmetry order and compactness – and exactly one ingredient remains that is not already FFGFT: the spiral-time. Its operators (R , K) depend on neither 3- nor 5-foldness and survive the switch to 3-fold unchanged. *What substantially remains of HLV is therefore not a geometry but a time* – and it is there, in the dynamics, that it meets FFGFT's non-Markovian kernel from Doc. 282. The geometric remainder – icosahedral 5-foldness – is an elegant but physically empty branch: no sector realises it, and φ is not specific (Doc. 276/281).

A short geometric lesson follows. Five-foldness is *non-crystallographic* – it does not tile periodically and forces decompactification; three-foldness is *crystallographic*, of the same family as six-foldness (the triangular/hexagonal lattice). Where geometry carries the mass structure it is therefore hexagonal, not pentagonal – and this is no new finding but FFGFT's own anchor: the *Euler Tonnetz* is a hexagonal lattice (Doc. 275, "the frame anchors in the Euler Tonnetz, not in the golden ratio").

The defensible form of the thesis

"Hexagonal" means the crystallographic family (3-/6-fold), not the claim that space-time is a hexagonal tiling – FFGFT's base remains T^4 with a Z_3 generation symmetry. Empirically 3-fold-confirmed so far are only the charged leptons; the hadrons are near-degenerate ($Q \approx 1/3$), hence neither 3- nor 5-fold. What stands is: *where geometry carries the mass structure it is hexagonal/crystallographic; the pentagonal stays an elegant, physically empty branch; and what substantially remains of HLV is its time, not its geometry.*

.11 Conclusion

All three parts of the criterion are answered. (1) Same object: a Z_3 circulant from HLV's own geometry. (2) Map: the C_3 -axis embedding of FFGFT's Z_3 into HLV's icosahedral group. (3) Divergence: fundamental, and in three layers – compactness (decompactified/metric vs compact/ $\tilde{T} \cdot m = 1$), symmetry order (5-fold φ , $1/\sqrt{5}$ vs 3-fold Koide, $\sqrt{2}$) and phase (HLV's dynamic spiral-time operator-phase R vs FFGFT's fixed scalar $2/9$; § 8). Compactness stays decisive for the *mass* scale (no quantised modes, no intrinsic m); HLV's *time*, however, sits

in the spiral-time, independently of it. No parameter of HLV reaches FFGFT's protected Koide point – yet at the time/memory level both non-Markovian kernels are of the same class: *a spatial fork, a possible temporal bridge* (§ 8).

This is exactly what the criterion is built to deliver: a *genuine* connection – shared Z_3 -circulant family, explicit C_3 map – *with* a real divergence point, and demonstrably *not* Stufe-0 triviality (the protected structure separates, only the meaningless part coincides). FFGFT and HLV sit at two *different* points of the same family. It is a *fork*, not a merge: a common stem (a Z_3 circulant from a higher-dimensional lattice with a hidden space), two branches according to the symmetry order.

For the correspondence the statement is precise and fair: *we meet in the Z_3 circulant along your C_3 axis; we separate because your icosahedral 5-fold symmetry rigidly forces $1/\sqrt{5}$, while FFGFT's Koide condition demands a 3-fold $\sqrt{2}$ – and no parameter of your construction bridges that.* Reproducible with `ffgft_hlv_c3_bridge_probe.py` (six checks, numpy-only).